

NYC 311 Service Requests & Operations Analysis

Descriptive analysis of a 500K-row NYC 311 service request sample, built in Power BI

1. Business Problem

The objective of this analysis is to understand how NYC 311 service requests are distributed across complaint types, boroughs, agencies, and channels, and to evaluate how efficiently requests are being resolved.

The analysis focuses on four key business questions:

- What types of problems generate the highest number of service requests?
- Which boroughs and agencies handle the largest workloads?
- How effectively are requests being closed?
- Which complaint types and locations have longer resolution times?

The analysis uses a 500K-row sample of NYC 311 service requests and was developed using **Power BI** for interactive analysis and dashboarding.

2. Key Performance Indicators

KPI	Result
Total Requests	500K
Closed Requests	428K
Open Requests	72K
Closure Rate	85.67%
Average Resolution Time	64.67 hours
Long-Running Requests (>72 hours)	88K
Long-Running Request Share	17.6%
Agencies Handling Requests	14

3. Key Insights

3.1 Demand is Concentrated Around a Few Complaint Types

Illegal Parking is the largest complaint category, with approximately **79K requests** and a **15.84% share** of total requests.

Other major complaint categories include:

- Noise – Residential
- Noise – Street/Sidewalk
- Blocked Driveway
- Unsanitary Condition
- Street Condition
- Water Maintenance

Insight: A relatively small number of complaint categories account for a substantial portion of service demand.

3.2 Brooklyn Has the Highest Request Volume

Brooklyn generated approximately **160K requests**, followed by:

- Queens — 127K
- Manhattan — 97K
- Bronx — 96K
- Staten Island — 20K

Insight: Service demand is not evenly distributed geographically. Brooklyn represents the largest workload among the boroughs shown in the dashboard.

3.3 Online Is the Dominant Request Channel

The channel analysis shows approximately:

- Online — 256K
- Phone — 108K
- Mobile — 95K
- Other/Unknown channels — smaller shares

Online requests account for approximately **51.17%** of the requests shown.

Insight: Digital channels represent an important part of the 311 service-request process.

3.4 NYPD Handles the Largest Agency Workload

NYPD handles approximately **233K requests**, substantially more than the next agencies in the workload chart.

The agency analysis also shows differences in:

- Request volume
- Closed requests
- Closure rates
- Average resolution hours

Insight: Agency workload is highly concentrated, making workload distribution and resolution performance important operational considerations.

3.5 Resolution Time Varies Considerably

The overall average resolution time is **64.67 hours**.

The borough analysis shows differences in average resolution time, with **Unspecified** and **Bronx** showing higher average resolution times than Queens.

The agency analysis also shows substantial variation in average resolution hours across agencies.

Insight: Resolution performance is not uniform across agencies or locations, indicating that workload, complaint characteristics, or operational processes may influence resolution time.

3.6 Long-Running Requests Require Attention

Approximately **88K requests**, or **17.6%**, take more than 72 hours to resolve.

The highest volumes of long-running requests are associated with:

- Brooklyn — 28.6K
- Bronx — 19.3K
- Manhattan — 18.4K
- Queens — 18.1K

The major complaint categories among long-running requests include:

- Unsanitary Condition — 13.3K
- Paint/Plaster — 5.9K
- Plumbing — 5.5K
- Water Leak — 4.7K
- Street Condition — 4.7K

Insight: Long-running requests are concentrated in specific boroughs and complaint categories, providing clear areas for operational monitoring.

4. Business Recommendations

Recommendation 1 — Prioritize High-Volume Complaint Categories

Use complaint-volume trends to prioritize operational resources around high-demand categories such as **Illegal Parking, Noise, Blocked Driveway, and Unsanitary Conditions**. This can help agencies focus capacity where the largest number of service requests originate.

Recommendation 2 — Monitor Long-Running Requests

Create an operational SLA monitoring process for requests exceeding **72 hours**. Requests approaching the 72-hour threshold could be flagged for review so that teams can identify potential delays before they become significantly overdue.

Recommendation 3 — Focus on High-Volume Boroughs

Brooklyn, Queens, Manhattan, and the Bronx account for the majority of the request volume shown in the analysis. Agencies could use borough-level workload monitoring to better align staffing and operational capacity with service demand.

Recommendation 4 — Investigate Agency-Level Resolution Differences

The dashboard shows significant differences in average resolution hours between agencies. Agencies with comparatively high resolution times should be examined further to understand whether the differences are associated with:

- Complaint complexity
- Workload
- Resource availability
- Operational processes
- Request characteristics

The dashboard identifies these areas for investigation but does not establish the underlying cause.

Recommendation 5 — Strengthen Digital Service Monitoring

Since Online is the largest request channel, digital request volumes should be monitored regularly. Tracking channel trends can help identify changes in how residents interact with the 311 system and support decisions around digital-service capacity.

5. Data Quality & Interpretation Notes

The dashboard is based on a **500K-row sample** of NYC 311 service requests.

The analysis should be interpreted as a descriptive analysis of the available dataset rather than a complete evaluation of NYC's entire 311 operation.

The agency closure-rate visualization contains some values above 100%. These should be treated as a **data/model validation point** rather than interpreted as agencies literally closing more requests than they received.

Possible areas for further validation include:

- Duplicate records
- Request status definitions
- Agency/request relationships
- Filter context
- Records with missing or unspecified agency information
- Calculation logic for agency-level closure rates

6. Final Business Takeaway

The analysis shows that NYC 311 demand is concentrated around a relatively small number of complaint categories and high-volume boroughs. While approximately **85.67% of the analyzed requests are closed**, around **88K requests take more than 72 hours** to resolve.

The analysis therefore highlights two major operational priorities:

- **Manage demand concentration** by monitoring high-volume complaint types and boroughs.
- **Improve resolution monitoring** by identifying long-running requests and investigating agencies or complaint categories associated with higher resolution times.

Overall, the dashboard transforms raw service-request data into an operational view of **demand, workload, closure performance, and resolution delays**.